

# Radial extension and dimension transfer

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## Scope

A surjective isometry between unit spheres produces a radial map and transfers finite-dimensionality between the ambient spaces. The radial map has the recorded metric bounds; it is not asserted to be linear or globally isometric.

7 direct Cards + 0 reused prerequisites = 7 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

## Cards in this route

- Key result: [Isometric unit spheres force equal finite dimensions](#)
- Key definition: [Radial extension of an isometry between unit spheres](#)

[Read this route with prerequisites](#)

## Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#)

## Level 0 (2 Cards)

Level 0

### Radial extension of an isometry between unit spheres

Extends a bijective sphere isometry to all vectors by retaining each radius and transporting its unit direction.

`MathlibAnnex.Sphere.radialExtension`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

### A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Continuous linear maps bounded by a seminorm](#)

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## Level 1 (3 Cards)

Level 1

### Continuous linear maps bounded by a seminorm

Defines the pointwise contraction condition relative to a stored equivalent seminorm.

`MathlibAnnex.EquivalentSeminorm.IsContraction`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

### The radial extension is 3-Lipschitz

Separates radial variation from the change of unit direction to obtain the global bound with coefficient  $1 + 2$ .

`MathlibAnnex.Sphere.lipschitzWith_radialExtension`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#)

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## Level 1

### The inverse sphere isometry gives the inverse radial map

Shows that extending the inverse sphere isometry radially undoes the forward radial extension.

`MathlibAnnex.Sphere.radialExtension_leftInverse`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#)

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## Level 2 (2 Cards)

## Level 2

### Isometric unit spheres force equal finite dimensions

Compares Hausdorff dimensions in both directions using the surjective Lipschitz radial maps.

`MathlibAnnex.Sphere.finrank_eq`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

## Level 2

### Finite-dimensionality transfers across a sphere isometry

Uses the radial homeomorphism to transfer local compactness, then applies the finite-dimensionality criterion for real normed spaces.

`MathlibAnnex.Sphere.finiteDimensional_codomain`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)